



SAVITRIBAI PHULE PUNE UNIVERSITY
JAYAWANTRAO SAWANT COLLEGE OF ENGINEERING ,HADAPSAR ,PUNE



Department of Computer Engineering

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Project Stage-I Review-I

**AI-Based Underwater Monitoring, Image Enhancement,
Detection and Deep Mapping.**

Final Year Of Computer Engineering

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Introduction

Underwater pollution is a growing environmental issue affecting marine life, water quality, and ecosystems. Traditional underwater monitoring is labor-intensive, risky, and inefficient. Our project proposes an AI-based solution that enhances underwater images, detects and classifies waste, estimates depth, and maps pollution zones in real-time. It aims to provide a smart, automated system that helps in monitoring and cleaning underwater areas more efficiently.

Manual cleaning of underwater environments is risky, slow, and expensive. Divers cannot always assess what lies beneath or how safe it is to clean certain zones. Moreover, without real-time data or visuals, cleanup authorities often guess where to start, which results in inefficient resource use and time loss.

Literature Survey(Explain your Reference Paper)

Sr. No.	Title	Year	Author	Remark	Gap Analysis
1.	Deep Learning Innovations for Underwater Waste Detection: An In-Depth Analysis	2023	Jaskaran Singh Walia, Pavithra L K Department of CSE, VIT Chennai	Highlights the growing importance of deep learning in identifying underwater waste with improved precision. Focuses on innovation in feature extraction and classification.	While innovation is discussed, real-time implementation, hardware constraints, and noisy underwater conditions are not deeply addressed. Limited experimentation with diverse datasets.
2.	Underwater Target Detection Using Deep Learning: Methodologies, Challenges, Applications, and Future Evolution	2023	Anwar Khan, Mostafa M. Fouda, Dinh-Thuan Do, Abdulaziz Almaleh, Abdullah M. Alqahtani, Atiq Ur Rahman	A comprehensive review covering deep learning methods (YOLO, CNNs),	The paper lacks real-world deployment and comparative model benchmarking. Also does not address power efficiency or model robustness in dynamic underwater conditions.

3	Underwater Object Detection using Image Enhancement and Deep Learning Models	2023	Adane Nega Tarekegn, Faouzi Alaya Cheikh, Mohib Ullah, Erdeniz Erol, et al.	Combines image enhancement (GANs, U-Net, WaterGAN) with object detection using YOLO and other deep learning models to improve clarity and accuracy	Though enhancement boosts detection, the impact on computational load and real-time feasibility is underexplored. No clear comparative analysis of enhancement vs. raw image detection performance.
4	Deep Learning Innovations for Underwater Waste Detection: An In-Depth Analysis	2025	J.S. Walia, K. Haridass, P.L.K. VIT Chennai	Focuses on DL-based waste detection using segmentation and classification.	High computational load; lacks comparison with raw detection.

Motivation

- Rising pollution in lakes, rivers, and coastal areas demands smart solutions.
- There's limited availability of real-time underwater AI monitoring systems.
- We are motivated to build a safe, fast, and effective AI-powered monitoring and mapping system.
- It contributes to environmental sustainability and supports clean water initiatives
- To build a smart, automated solution that reduces human risk while increasing the effectiveness of trash detection and removal
- Manual underwater garbage removal is dangerous, inefficient, and often ineffective due to limited visibility, unknown water depth, and the uncertain type of waste present. Human divers face risks from contaminated water, sharp materials, or entanglement in underwater debris.

Problem Statement

To design and implement a real-time AI system that enhances underwater images, detects and classifies trash, estimates the depth and density of pollution, and categorizes underwater zones based on risk level. The system must be capable of working in real-time scenarios using camera feeds or videos and should present this information on a user-friendly dashboard. The goal is to reduce manual effort, improve safety, and support environmentally responsible underwater waste management.

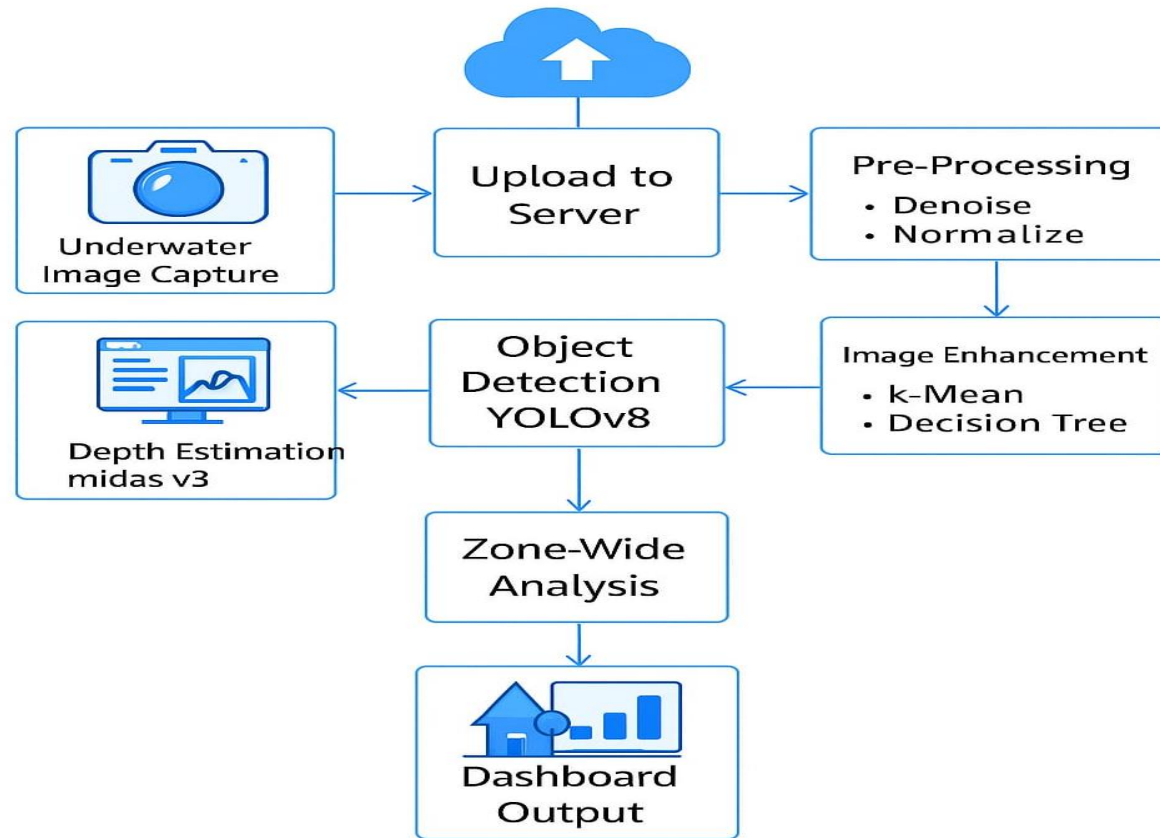
Objective

- To develop a system that captures underwater images or videos using real-time or offline sources (e.g., cameras, recorded footage).
- To enhance underwater images using deep learning models like U-Net, Water-Net, or GAN for better clarity and contrast.
- To estimate the depth of each object in the image using MiDaS v3.1 (monocular depth estimation) without requiring special hardware.
- To calculate trash density in each frame based on the percentage of area covered by detected waste.
- To create rules or thresholds to label zones as Clean, Moderate, or Critical based on trash density and depth information.
- To visualize all the processed data (original frame, enhanced frame, detected objects, depth map, etc.) on a single interactive dashboard.

Scope

- Applicable in rivers, lakes, ponds, and shallow oceans.
- Helps municipal bodies, NGOs, and government teams plan cleanups.
- Can be scaled to work with drones, AUVs, or underwater robots.
- Supports real-time and offline data processing.
- Can be extended to detect marine life, cracks in infrastructure, or pipeline monitoring in the future.

Architecture Diagram



Conclusion

This project successfully combines deep learning and computer vision to build a real-time underwater monitoring system. It detects, classifies, and maps underwater garbage and provides valuable insights through a dashboard. The system not only improves safety and efficiency but also promotes sustainability. It can revolutionize how authorities and organizations approach underwater cleaning and monitoring tasks.

References

- [1]Jaskaran Singh Walia, Pavithra L K Department of CSE, VIT Chennai (2023)Deep Learning Innovations for Underwater Waste Detection: An In-Depth Analysis
- [2] Anwar Khan, Mostafa M. Fouda, Dinh-Thuan Do, Abdulaziz Almaleh, Abdullah M. Alqahtani, Atiq Ur Rahman(2023) Underwater Target Detection Using Deep Learning: Methodologies, Challenges, Applications, and Future Evolution
- [3] Adane Nega Tarekegn, Faouzi Alaya Cheikh, Mohib Ullah, Erdeniz Erol, et al.(2023) Underwater Object Detection using Image Enhancement and Deep Learning Models
- [4] J.S. Walia, K. Haridass, P.L.K.VIT Chennai(2025)Deep Learning Innovations for Underwater Waste Detection: An In-Depth Analysis

THANK YOU
